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Course Description

6th Grade STEM is an introduction to Earth and Space Science. It is designed to unite our world view as citizens of the planet, nurture our understanding of how the Earth's systems are interrelated and inspire the next generation of scientists and engineers.

The year begins with a short introduction to the Scientific Method and the Engineering Design Process as well as the principal tools and techniques of a scientist and engineer. Then it's time to leave this planet we call home and explore the wonders of the Universe. We will return to the "Big Bang" and discover the universe's beginnings. After the Winter Break, we will be going back in deep time, 4.5 billion years ago to discover how the Earth was made and learn the key events that shaped our world. After Spring Break, we will conclude the year with an investigation of two pressing global issues, climate change and the low energy future.

2025 – 2026 Scope and Sequence: August-June

All Unit Dates are *projected* and may be subject to change.

Unit 1: Why STEM

5 Aug – 30 Aug

Key Concept: Systems

Related Concept(s): Patterns, Form, Function

Global Context: Science and Technical Innovation

Inquiry statement: The Scientific Method and the Engineering Design Process allow us to investigate the form and function of patterns observed in natural systems or designs.

Inquiry questions:

Factual: How do scientists and engineers use the Scientific Method and the Engineering Design Process?

Conceptual: How does experimental science and engineering expand our body of knowledge?

Debatable: Can the Scientific Method or the Engineering Design Process be used to answer any question or solve any problem?

Content

- **Know:** Scientific Method, Engineering Design Process, Extinction Level Events
- **Understand:** How the Scientific Method is used to answer questions. The linkage between ELE's and mass extinctions.
- **Do:** Lab Safety, Scientific Method/Engineering Design Process, variables and hypothesis formulation, Lab report format.

Summative Assessment Tasks

- ELE Poster (Crit. A/D)

Unit 2: Our Universe

2 Sep – 18 Dec

Key Concept: Change

Related Concepts(s): Models, Interaction, Movement

Global Context: Science and Technical Innovation

Inquiry statement: Models allow scientists to understand the movements and interactions that change our Universe.

Inquiry questions:

Factual: What forces govern our Universe?

Conceptual: How was our Universe created?

Debatable: How old is our Universe?

Content

- **Know:** Structure and states of matter, types of objects in the universe, how we use models to help us understand our Universe.
- **Understand:** How the Universe was formed and that it is still expanding, formation of stars, importance of gravity, scale of the Universe.
- **Do:** Mass Lab, Volume Lab, Density Lab, Alien in Me, Great Pluto Debate.

Summative Assessment Tasks

Density Lab (Crit. B/C)

Great Pluto Debate (Crit. A/D)

Unit 3: How the Earth was Made

6 Jan - 13 Mar

Key concept: Systems

Related concepts(s): Evidence, Environment

Global context: Orientation in Space and Time

Inquiry statement: Evidence allows us to see that the interplay of geologic, environmental, and biological systems created our Earth.

Inquiry questions:

Factual: Where is the evidence of the 4.5 billion year development of the Earth?

Conceptual: How do we know the age of our Earth?

Debatable: What are the key events that resulted in our world being what it is today?

Content

- **Know:** 25 key events in the 4.5 billion year history of our Earth.
- **Understand:** How the key events shaped our Earth into what we see today.
- **Do:** Explore the evidence in the rock, investigate how our Earth changed over the past 4.5 billion years, and understand what fossils tell us about life in the deep past. Rock Lab, Fossil Lab.

Summative Assessment Tasks

Ages of the Earth Presentation (Crit A/D)

The Great Oxygenation Lab (Crit. B/C)

Earth Timeline Project (Crit. A/D)

Key Concept: Change**Related Concept(s):** Balance, Energy, Consequences**Global Context:** Globalization and Sustainability

Inquiry statement: Exploring mankind's use of energy and its link to our climate will reveal a delicate balance of forces that are changing and will have significant consequences for the sustainability of life on our planet.

Inquiry questions:

Factual: What factors affect climate?

Conceptual: How is mankind's use of energy changing climate factors?

Conceptual: How will a low-energy future affect our ability to deal with climate change?

Debatable: Is human activity having a significant impact on climate change?

Content

- **Know:** The evidence behind Climate Change and the Low Energy future.
- **Understand:** How the factors affecting climate change are linked to energy usage.
- **Do:** Identify the difference between weather and climate, explore the factors that are causing Earth's climate to change, investigate the impact of those changes on our Earth, identify where the energy comes from and how it is used, explore trends in energy use growth and its implications for the future, investigate the nexus created by the changing climate and the potential to run out of energy in the near future.

Summative Assessment Tasks

Critical Lab Skills Test (Crit. B/C)

Sources of Energy Presentation (Crit. A/D)

Baseball Projects (Crit. A/D)

STEM Assessment Criteria

	Achievement Levels
A. Knowing and Understanding	0-8
B. Inquiring and Design (1 st half of Experimental/Design Cycle)	0-8
C. Processing and Evaluation (2 nd half of Experimental/Design Cycle)	0-8
D. Reflecting on the Impacts of Science/Engineering	0-8

Student activities will be assessed against the criteria listed above. End of quarter letter grades will be assigned based on student learning progress over the quarter.

Contact Information: email - deachus@starofthesea.org

- Parents with questions or concerns may directly contact any instructor.
- Students are encouraged to use Flex Time or lunch recess to obtain additional learning support. In addition, **the Innovations Lab will be open each Wednesday afternoon (except Mini-days) until 4 pm for any student needing help or to finish lab work.**
- Students and/or parents should use email to contact their instructor to make arrangements for additional learning opportunities.